

Yash Kherwal

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EDUCATION

Indian Institute of Technology Roorkee

2022 – 2026

B.Tech in Chemical Engineering

EXPERIENCE

AI Engineer Intern

Feb 2026 – Apr 2026

Orange Labs — EdTech ERP platform for schools and colleges

- Built **CCTV-based automated attendance pipeline** — computer vision model detects and identifies students from live camera feeds, auto-syncing presence records into the school ERP; eliminated manual marking and enabled real-time classroom tracking.
- Engineered **AI-generated personalised lecture pipeline**: ingested real teacher audio/video as training data to synthesise **teacher-cloned video lectures** — students receive curriculum content delivered in their own school teacher's voice and likeness, improving engagement over generic video content.

Software Development Intern

May 2025 – June 2025

ScoreMe Solutions

- Designed a **two-stage ML classification pipeline** for fintech PDF processing — rule-based heuristics handle deterministic inputs first; **Random Forest classifier** resolves ambiguous cases, avoiding unnecessary model inference on clear-cut documents and reducing classification latency.
- Implemented **confidence calibration layer**: predictions below threshold withheld from downstream **credit scoring pipelines** and flagged for analyst review — prevents silent misclassification propagation; integrated **Tesseract OCR** via **PDFBox** as a fallback parsing path for scanned inputs, extending pipeline coverage without adding routing complexity.

Engineering Intern

June 2025 – July 2025

Bharat Electronics Limited

- Implemented a **binary stream parser** for **ASTERIX CAT048** radar protocol per EUROCONTROL spec — decoded variable-length optional-field packet structures into normalised aviation records; designed **concurrent ingestion** with **backpressure handling** to maintain throughput and prevent packet loss under sustained high-frequency surveillance loads.

PROJECTS

Autonomous Data Analysis Workflow | *Python, LangGraph, CrewAI, n8n, MCP, FastAPI, Docker, AWS EC2*

- Designed a **stateful 5-agent LangGraph workflow** (Query Planner → Code Writer → Executor → Error Fixer → Insight Generator); Query Planner resolves **intent ambiguity** and enforces **deterministic routing** to SQL or Python execution paths before generation, eliminating downstream type errors.
- **Error Fixer agent** implements **autonomous exception recovery**: intercepts runtime failures, classifies error type, regenerates targeted fix, retries execution — structured escalation to operator only on unresolvable failures, reducing noise in production workflows.
- **n8n** decouples workflow orchestration from agent logic; **MCP** exposes tools (files, DBs, APIs) as standardised context — Executor runs in **Docker sandbox** to isolate LLM-generated code from host; deployed **FastAPI** backend on **AWS EC2**.

Hybrid RAG Retrieval System | *Python, LangChain, BGE-M3, AstraDB, FastAPI, AWS EC2*

- Diagnosed keyword retrieval failure on a 500+ document corpus — BM25 missed semantically relevant results due to vocabulary mismatch; introduced **hybrid retrieval** combining **BGE-M3** dense embeddings with sparse BM25 scores, improving recall on paraphrased and domain-specific queries.
- Designed **decoupled ingestion and serving**: offline embedding pipeline feeds **AstraDB** independently of the **FastAPI serving layer** on **AWS EC2** — corpus updates without serving downtime; **response groundedness** enforced via prompt constraints to prevent hallucination on out-of-context queries.

Semantic Document Search | *Python, LangChain, BGE-M3, AstraDB, FastAPI, AWS EC2*

- Built a **hybrid retrieval system** combining **BGE-M3 embeddings** and **BM25 scoring** to improve semantic search quality across 500+ documents while preserving exact-match retrieval performance.

YAAR — AI Behavioral Companion | *React Native, LLM APIs, FastAPI, Persistent Memory*

- Designed a **persistent personality memory system**: each user interaction updates a **personality metric vector** tracking dimensions (humour tolerance, self-awareness index, reaction intensity) — stored per-session and injected into the **LLM context window** at inference time to steer tone and content without fine-tuning.

TECHNICAL SKILLS

LLM & Generative AI: LangGraph, LangChain, CrewAI, MCP, Agentic Workflows, Structured Outputs, Prompt Engineering, Voice/Video Synthesis

Computer Vision: OpenCV, Object Detection, Face Recognition, Real-Time Video Processing, CCTV Analytics

Retrieval: Hybrid RAG, BGE-M3, Dense + Sparse Retrieval, AstraDB, Vector Databases, ANN Indexing

ML / AI: TensorFlow, Scikit-learn, Multi-Agent Orchestration, n8n, Inference Optimization

Backend & Infra: FastAPI, Docker, AWS EC2, REST APIs, Stream Processing

Languages & Data: Python, SQL (window functions, cohort queries), Java, C++, DSA, Pandas, pdfplumber